

1. Administrative & Study Identification

Study Title: A Study of Secukinumab to Evaluate Maintenance of Response in Participants With Non-radiographic Axial Spondyloarthritis Who Achieved Remission **Official Title:** A Multicenter Study of Secukinumab, With a Randomized Double-blind, Placebo-controlled Withdrawal-retreatment Period, to Evaluate Maintenance of Response in Participants With Non-radiographic Axial Spondyloarthritis Who Achieved Remission **Short Title/ Acronym:** Secukinumab nr-axSpA Remission Maintenance Study **Protocol Number:** CAIN457I2401 **NCT Number:** NCT05622708 **Trial Status:** RECRUITING (Currently recruiting participants) **Sponsor/ Company Name:** Novartis Pharmaceuticals (Novartis) **Investigational Product:** Secukinumab **Phase of Development:** Phase IV (PHASE4) **Criteria Type:** PHASE_REQUIREMENT **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the maintenance of remission on continued secukinumab treatment compared to placebo using a randomized withdrawal design, specifically assessing the proportion of participants remaining flare-free at Week 120. **Secondary Objectives:** To evaluate the continuous maintenance of Ankylosing Spondylitis Disease Activity Score - C-reactive protein (ASDAS-CRP) Inactive Disease (ID) response, the time to disease flare, and the number of participants with Adverse Events.

2.2 Background & Rationale To address the clinical question of whether prolonged chronic dosing with secukinumab is necessary for participants with non-radiographic axial spondyloarthritis (nr-axSpA) who have successfully achieved remission. Assessing if treatment withdrawal leads to flare will help guide long-term therapeutic strategies and optimize the use of biologic therapies in this specific patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (Randomized withdrawal design) **Blinding:** Double-blind **Randomization:** Randomized withdrawal and retreatment

3.2 Planned Enrollment & Duration Target **\$N\$:** 340 evaluable patients **Number of Sites:** Multicenter (Global, including Belgium

and 17 others) **Estimated Study Start:** 27-Mar-2023 **Estimated Primary Completion:** 20-Jun-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or non-pregnant, non-lactating female participants, at least 18 years of age (up to 100 years). 2. Clinical diagnosis of axSpA AND according to ASAS axSpA criteria (Inflammatory back pain $\geq$ 6 months, onset before 45 years of age, Sacroiliitis on MRI with $\geq$ 1 SpA feature OR HLA-B-27 positive with $\geq$ 2 SpA features). 3. Objective signs of inflammation at screening, evident by MRI with Sacroiliac Joint inflammation AND/OR hsCRP > ULN. 4. Active nr-axSpA as assessed by total BASDAI $\geq$ 4 cm (0-10 cm), spinal pain $\geq$ 4 cm, and total back pain VAS $\geq$ 40 mm at baseline. 5. Participants who have been on at least 2 different NSAIDs at the highest recommended dose for at least 4 weeks in total prior to baseline with an inadequate response, failure to respond, or intolerance.

4.2 Exclusion Criteria 1. Radiographic evidence for sacroiliitis, grade $\geq$ 2\$ bilaterally or grade $\geq$ 3\$ unilaterally (radiological criterion according to the modified New York diagnostic criteria for AS). 2. Participants taking high-potency opioid analgesics (e.g., methadone, hydromorphone, morphine). 3. Active ongoing inflammatory diseases other than nr-axSpA (e.g., active inflammatory bowel disease, uveitis) or history of ongoing, chronic, or recurrent infectious disease (e.g., tuberculosis). 4. Previous exposure to secukinumab or any other biologic drug directly targeting IL-17 or IL-17 receptor, or prior immunomodulatory biologic agents including those targeting TNF α .

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Name: Secukinumab (Preferred Name: secukinumab) **Trade Name:** Cosentyx **ATC Code:** L04AC10 **Dosage/Frequency:** * Open-label Secukinumab PFS (prefilled syringe) will be labeled as AIN457 150mg/1mL. * Double-blind Secukinumab and Placebo PFS will be labeled as AIN457 150mg/1mL/Placebo. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** Up to 120 weeks with the last treatment administration at Week 116. Total study duration is up to 128 weeks from Baseline. **EMA URL:** [Cosentyx EPAR Product Information](#)

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of non-steroidal anti-inflammatory drugs (NSAIDs), as specified by baseline eligibility. **Prohibited:** High potency opioid analgesics or concurrent use of other biologic immunomodulators during the trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline BASDAI, VAS, and CRP assessments
Baseline	Day 0	Initial open-label dosing to achieve remission, baseline safety evaluations
Treatment Period 1	~ Every 12 weeks	Open-label site visits (First visit ~1 month after Baseline, then ~every 12 weeks)
Interim / TP 2	~ Every 4 weeks	Evaluation of remission (ASDAS-CRP < 1.3), Randomization to withdrawal (Placebo) or maintenance (Secukinumab); visits every 4 weeks
End of Study	Week 128	Primary endpoint evaluation (Flare-free assessment at Week 120), Final Vital Signs, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants in the randomized withdrawal population remaining flare-free during Treatment Period 2 (at Week 120). A flare is defined as ASDAS-CRP ≥ 2.1 at 2 consecutive visits, or ASDAS-CRP > 3.5 at any visit during Treatment Period 2, starting at Week 60. **Measurement Method:** Clinical assessment utilizing standardized disease activity indices and patient-reported outcomes (Spinal pain, Patient global assessment, Peripheral pain/swelling, morning stiffness duration, CRP).

7.2 Secondary & Exploratory Endpoints 1. Time to flare during Treatment Period 2. 2. Number of participants with Adverse Events. 3. Maintenance of remission defined as Ankylosing Spondylitis Disease Activity Score - C-reactive protein (ASDAS-CRP) Inactive Disease (ID) response (ASDAS-CRP < 1.3).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient reporting, laboratory evaluations, and HCP assessments at scheduled visits. **Serious Adverse Events (SAEs):** Reported to the sponsor and regulatory authorities within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External independent DMC established by Novartis to oversee safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Set for adverse events. **Sample Size Justification:** $N=340$ appropriately powered to demonstrate a statistically significant difference in flare rates between the continuous secukinumab arm and the placebo withdrawal arm.

Handling of Missing Data: Non-responder imputation (NRI) for categorical efficacy endpoints; mixed-effects models for repeated measures (MMRM) for continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Secure electronic data capture (EDC) system with comprehensive audit trails and data clarification processes.

11. Study Identifiers & Governance

Primary ID: CAIN457I2401 **Posting ID:** 309611177861007345 **Registry Number:** NCT05622708 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Study of Secukinumab to Evaluate Maintenance of Response in Participants With Non-radiographic Axial Spondyloarthritis Who Achieved Remission **Source Level:** 5

12. Clinical Framework (nr-axSpA Specific)

Primary Condition / Disease Area: Non-radiographic Axial Spondyloarthritis (nr-axSpA) **Preferred UMLS Name:** Non-radiographic axial spondyloarthritis **Semantic Type:** Disease or Syndrome **Definition:** Chronic inflammatory conditions affecting the axial joints which cannot be detectable on x-rays. It is characterized by pain, stiffness of joints and inflammation. Non-radiographic axial spondyloarthritis can have symptoms onset before the age of 45 and progress to more severe ANKYLOSING SPONDYLITIS over time. **MeSH Code:** D000089202 **SNOMED ID:** 713777005 **Diagnosis Threshold:** Active nr-axSpA (BASDAI ≥ 4 cm) and no radiographic sacroiliitis **Medical Methodology:** Biologic Disease-Modifying Antirheumatic Drug (bDMARD) Therapy **Optimization Target:** Sustained Remission (ASDAS-CRP < 1.3)

13. Study Procedures & Methodology

Management Pathway: Treat-to-target followed by randomized withdrawal and potential retreatment **Education Protocol (HCP):** Protocol-specific training for investigators on withdrawal criteria and flare identification **Education Protocol (Patient):** Onsite training/study instructions regarding flare-reporting and medication administration **Quality Audit Mechanism:** Patient adherence monitoring and central imaging reader quality checks

14. Observation & Follow-Up Schedule

Total Duration: Up to 128 Weeks from Baseline (Treatment duration up to 120 weeks) **Onsite Visit Cadence:** Every 12 weeks in Treatment Period 1; Every 4 weeks in Treatment Period 2. **Remote Monitoring Frequency:** Supplemental remote contacts as defined by the clinical monitoring plan **Checklist Review Interval:** ePROs and symptom diaries evaluated at scheduled protocol assessments

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					